

Children and adolescents

Dosing in children with impaired renal and or hepatic function has not been studied.

Mode of Administration:
Intravenous

Contraindication:

Ificpro is contraindicated in individuals with a history of hypersensitivity to Ciprofloxacin or any other quinolone derivative. Its use is not recommended in children below the age of 12 years.

Warning and Precautions:

Warning:

The use of ciprofloxacin solution for infusion should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with ciprofloxacin solution for infusion should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Psychiatric reactions

Psychiatric reactions may occur even after the first administration of fluoroquinolones, including Ciprofloxacin. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or develops these reactions, Ciprofloxacin should be discontinued and appropriate measures instituted. Caution is recommended if Ciprofloxacin is to be used in psychotic patients or in patients with a history of psychiatric disease

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ificpro Injection 200mg/100ml should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with cortico-steroids'. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with Ificpro Injection 200mg/100ml should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilization). Corticosteroids should not be used if signs of tendinopathy occur.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paresthesia, hypesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition.

Precautions:

As Ificpro may cause CNS stimulation. It should be used with caution in patients with CNS disorders such as severe cerebral arteriosclerosis or epilepsy. Patients receiving this drug should be well hydrated to prevent crystalluria. Excessive alkalinization of urine should be avoided. The dosage should be reduced in patients with renal impairment. Reproduction studies in animals at doses up to 6 times the usual daily human dose have not revealed any evidence of impaired fertility or teratogenicity due to Ificpro. However, information from well-controlled studies in pregnant women is not available. Since Ificpro causes arthropathy in immature animals, It should not be used in pregnant and nursing women.

Interactions with Other Medicaments:

Serum concentrations and elimination half life of theophylline maybe increased when it is used concurrently with Ificpro. It is recommended that patients monitored for the signs of theophylline toxicity during concurrent use and dosage adjustments made appropriate. Probenecid delays excretion of Ificpro.

Pregnancy and Lactation:

The safety and efficacy of ciprofloxacin in children, adolescents, pregnant women and lactating women have not been established.

Side Effects:

Ificpro (Ciprofloxacin) is generally well tolerated. During clinical trials in a large number of patients, adverse effects related to drug occurred infrequently and were commonly reported as diarrhoea, vomiting, abdominal pain, headache, restlessness and rash. Other side effects which have been reported very rarely include arthralgia and increases in serum transaminases levels.

Musculoskeletal and connective tissue disorders*

Nervous system disorders*

General disorders and administrative site conditions*

Eye disorders*

Ear and labyrinth disorders*

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

Psychiatric disorders

Rare: Depression (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Very Rare: Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Exacerbation of Myasthenia Gravis

Post Marketing Experience

Instructions for Use:

Ificpro Injection 200mg/100m1 should be administered by IV infusion over a period of 60 minutes. Slow infusion into a large vein will minimize patient discomfort and reduce the risk of venous irritation. The infusion solution can be infused either directly or after mixing with other compatible infusion solutions.

Incompatibilities: Ificpro Injection 200m0/100m1 solution is compatible with physiological saline. Ringer solution and Ringer lactate solution, 5% and 10% Glucose solutions, 10% fructose solution, and 5% glucose solution with 0.225% Sodium chloride 0,0.45% Sodium Chloride. When Ificpro Injection 200mg/100m1 is mixed with compatible infusion solutions, for microbiological reasons and light sensitivity these solutions should be administered shortly after admixture.

Unless compatibility with other infusion solutions/drugs has been confirmed, the infusion solution must always be administered separately. The visual signs of incompatibility am eg., precipitation, clouding and discoloration.

Incompatibility appears with all infusion solutions/drugs that are physically or chemically unstable at the pH of the solution (eg. penicillins. heparin solutions). Especially on combination with solutions adjusted to an alkaline pH (pH of the ciprofloxacin infusion solutions: 3.9-4.5).

Symptoms and Treatment of Overdosage:

Based on the limited information available in two cases of ingestion stover 18g of ciprofloxacin, reversible renal toxicity has occurred. Therefore, apart from routine emergency measures, it is recommended to monitor renal function, including urinary pH and acidify, if required, to prevent crystalluria. Patients must be kept well hydrated, and in the case of renal damage resulting in prolonged oliguria, dialysis should be initiated. Serum levels of ciprofloxacin are reduced by dialysis.

Storage Conditions:

Store at temperature not exceeding 30°C. Protect from light. Do not freeze.

Shelf Life:

3 years from the month of manufacturer

Therapeutically Code (if any):

J01MA02: CIPROFLOXACIN

Presentation:

Injection is available in a pack of 100ml transparent, colorless plastic bottles.

Date of Revision: September 2024



Manufactured by:
UNIQUE PHARMACEUTICAL LABS.
(A Div. of J. B. Chemicals & Pharmaceuticals Ltd.)
Plot No. 304- 308, G.I.D.C Industrial Area, Panoli 394 116, Gujarat State, India.
* Trade Mark

Imported by:
Unimed Sdn Bhd
53, Jalan Tembaga SD5/2B, Bandar Sri Damansara 52200,
Kuala Lumpur, Malaysia.

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For the use of Medical Practitioner or a Hospital or a Laboratory.

Ificpro* Injection 200mg/100ml (Ciprofloxacin Injection USP)

Composition:

Each 100ml contains Ciprofloxacin Hydrochloride USP equivalent to Ciprofloxacin 200mg

Product Description:

A clear, colourless to pale yellow liquid.

Pharmacodynamics:

Ciprofloxacin is from the quinolone group; these substances are also known as gyrase inhibitors.

Microbiology:

Gram-Negative Organisms

Enterobacteriaceae including *E.coli*, *Salmonella species*, *Klebsiella species*, *Shigella*, *Proteus mirabilis*, *Proteus vulgaris*, *K. oxytoca*, *Yersinia enterocolitica*, *Enterobacter*, *Citrobacter*, *Morganella morganii*, *Pseudomonas aeruginosa*, *Hemophilus Influenza*, *Acinetobacter*, *Campylobacter*, *Brucella melitensis*, *Pasteurella multocida*, *Eikenella corrodens*, *Flavobacterium*, *Moraxella*, *Gardnerella Vaginalis*, *Legionella species*, *Vibrio cholerae* and *Vibrio parahemolyticus*, *Neisseria meningitidis* and *Neisseria gonorrhoea* including b-lactomase producing strains.

Gram-Positive Organisms

Staphylococcus aureus including b-lactamase producing and methicillin-resistant strains, *streptococcus pneumoniae* group A b-hemolytic streptococci, group B streptococci and other streptococci, *Enterococci* including *Enterococcus faecalis*, *Corynebacterium* and *Listeria monocytogenes*.

Other microorganisms

Anaerobic bacteria including actinomyces, Bifidobacterium, Peptococcus, Clostridium perfringens, Eubacterium, Propioribacterium acres, Veillonella and some strains of Bacteroides, Chlamydia, Mycoplasma and Mycobacterium. Ificpro is effective against organisms resistant to nalidixic acid. Since cross-resistance is unlikely, Ificpro may be used to treat infections caused by organism resistant to other class of antibacterial such as aminoglycosides, penicillin, sulfonamides, tetracyclines and cephalosporins.

Pharmacokinetics:

The pharmacokinetic of ciprofloxacin are linear over the dose range of 200-400mg administered intravenously. The serum elimination half-life is approximately 5-6 hours and the total clearance is around 35L/hr. Comparison of the pharmacokinetic parameters following the 1st and 5th IV dose on a q12h regime indicates no evidence of drug accumulation.

After intravenous administration, approximately 50-70% of the dose is excreted in the urine as unchanged drug. Following a 200mg IV dose, concentrations in the urine exceed 200 µg/ml 0-2 hours after dosing and are generally greater than 15 µg/ml 8-12 hours after dosing. Following a 400mg IV dose, urine concentrations generally exceed 400 µg/ml 0-2 hours after dosing and are usually greater than 30 µg/ml 8-12 hours after dosing. The renal clearance is approximately 22 L/hr. The urinary excretion of ciprofloxacin is virtually complete by 24 hours after dosing.

The serum concentrations of ciprofloxacin and metronidazole were not altered when these two drugs were given concomitantly.

Co-administration of probenecid with ciprofloxacin results in about 50% reduction in the ciprofloxacin renal clearance and a 50% increase in its concentration in the systemic circulation. Although bile concentrations of ciprofloxacin are several fold higher than serum concentrations after intravenous dosing, only a small amount of the administered dose (<1%) is recovered from the bile as unchanged drug. Approximately 15% of an IV dose is recovered from the feces within 5 days after dosing.

After IV administration, three metabolites of ciprofloxacin have been identified in human urine which together account for approximately 10% of the intravenous dose. Pharmacokinetic studies of the oral (single dose) and intravenous (single and multiple dose) forms of ciprofloxacin indicate that plasma concentration of ciprofloxacin are higher in elderly subjects (> 65 years) as compared to young adults. Although the C max is increased 16-40%, the increase in mean AUC is approximately 30% and can be at least partially attributed to decreased renal clearance in the elderly. Elimination half-life is only slightly (~20%) prolonged in the elderly. These differences are not considered clinically significant.

In patients with reduced renal function, the half-life of ciprofloxacin is slightly prolonger and dosage adjustments may be required. In preliminary studies in patients with stable, chronic liver cirrhosis, no significant changes in ciprofloxacin pharmacokinetics have been observed. However, the kinetics of ciprofloxacin in patients with acute hepatic insufficiency have not been fully elucidated.

Following infusion of 400mg IV Ciprofloxacin every 8 hours in combination with 50mg/kg IV piperacillin sodium every 4 hours, mean serum ciprofloxacin concentrations were 3.02 µg/ml ½ hour and 1.18 µg/ml between 6-8 hours after the end of infusion

The binding of ciprofloxacin to serum proteins is 20 to 40%. After intravenous administration, ciprofloxacin is present in saliva, nasal and bronchial secretions, sputum, skin blister fluid, lymph, peritoneal fluid, bile, and prostatic secretions. It has also been detected in the lung, skin, fat, muscle, cartilage, and bone. Although the drug diffuses into cerebrospinal (CSF), CSF concentrations are generally less than 10% of peak serum concentrations. Levels of the drug in the aqueous and vitreous chambers of the eye are lower than in serum.

Indication(s):

Consideration should be given to available official guidance on the appropriate use of antibacterial agents.

For Ciprofloxacin solution for infusion:

Uncomplicated and complicated infections caused by ciprofloxacin susceptible pathogens.

- Infections of the respiratory tract:

Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by Klebsiella, Enterobacter, Proteus, E. Coli, Pseudomonas, Haemophilus, Moraxella catarrhalis, Legionella and Staphylococcus.

- Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis), especially if these are caused by Gram-negative organisms including Pseudomonas aeruginosa or by staphylococci.

- Infections of the eyes

- Infections of the kidneys and/or the efferent urinary tract Infections of the genital organs, including adnexitis, gonorrhea, prostatitis

- Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)

- Infections of the skin and soft tissue

- Infections of the bones and joints

- Sepsis

- Infections or imminent risk of infection (prophylaxis) in patients whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)

Dosage and method of administration

Dosage regimen

Unless otherwise prescribed, the following daily doses are recommended for adults

Table 1: Recommended daily doses of ciprofloxacin in adults

Indications	Daily dose of ciprofloxacin in mg for ciprofloxacin solution for infusion
Infections of the respiratory tract (according to severity and organism)	2 x 400 mg to 3 x 400 mg
Urinary tract infections: - acute, uncomplicated - cystitis in women (before menopause) - complicated	2 x 200 mg to 2 x 400 mg N/A 2 x 400 mg to 3 x 400mg
Genital infections - uncomplicated gonorrhoea (including extragenital sites of infection) - adnexitis, prostatitis, epididymo-orchitis	N/A 2 x 400 mg to 3 x 400 mg
Diarrhea	2 x 400mg
Other infections (See indications)	2 x 400 mg
Particularly severe, life-threatening infections, i.e. - Recurrent infections in cystic fibrosis - Bone and joint Infections - Septicaemia - Peritonitis. In particular when <i>Pseudomonas</i> , <i>Staphylococcus</i> or <i>Streptococcus</i> is present	3 x 400mg
Inhalational anthrax (post exposure)	N/A
Prophylaxis of invasive infections due to <i>Neisseria meningitidis</i>	N/A

Children and adolescents

Table 2: Recommended daily doses of ciprofloxacin in children and adolescents

Indications	Daily dose of ciprofloxacin in mg for ciprofloxacin intravenous
Infections in cystic fibrosis	3 x 10 mg/kg body weight (maximum of 400 mg per dose)
Complicated urinary tract infections and pyelonephritis	3 x 6 mg/kg body weight to 3 x 10 mg/kg body weight (maximum of 400 mg per dose)
Inhalational anthrax (post-exposure)	N/A

Missed dose

If a dose is missed, it should be taken as soon as the patient remembers and then treatment should be continued as prescribed. Double doses should not be taken to compensate for a missed dose.

Additional information on special patient population

Geriatric patients (> 65 years)

Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance. (See also 'Patients with renal and hepatic impairment')

Patients with renal impairment

Table 3: Recommended doses for patients with renal impairment

Creatinine Clearance (mL/min/ 1.73m ²)	Serum Creatinine [mg/100mL1	Total daily intravenous dose of ciprofloxacin
30 to 60	1.4 to 1.9	maximum 800mg
below 30	12.0	maximum 400mg

Patients with renal impairment on hemodialysis

For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100 mL, the maximum daily dose of ciprofloxacin should be 800 mg for an intravenous regimen. For patients with creatinine clearance less than 30 mL/min/1.23m, (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily dose of ciprofloxacin should be 400 mg for an intravenous regimen on dialysis days after dialysis.

Patients with renal impairment on continuous ambulatory Peritoneal dialysis (CAPD)

Addition of ciprofloxacin solution for infusion to the dialysate (intraperitoneal): 50 mg ciprofloxacin / liter dialysate administered 4 times a day every 6 hours.

Patients with hepatic impairment

in patients with impaired hepatic function no dose adjustment is required.

Patients with renal and hepatic impairment

For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal Impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100mL, the maximum daily dose of ciprofloxacin should be 800 mg for an intravenous regimen. For patients with creatinine clearance less than 30mL/min/1.73m² (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily dose of ciprofloxacin should be 400 mg for an intravenous regimen on dialysis days after dialysis